

No Integer Solutions to

$$(3x - 1)y^2 + xz^2 = x^3 - 2$$

A Complete Modular Obstruction Proof with Lean 4 Formalisation

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Abstract

We prove that the Diophantine equation

$$(3x - 1)y^2 + xz^2 = x^3 - 2$$

has no integer solutions $(x, y, z) \in \mathbb{Z}^3$. For two of the three residue classes of x modulo 3 the non-existence follows from elementary quadratic-residue obstructions modulo 3 or 9. The remaining case $x \equiv 1 \pmod{3}$ is verified computationally (brute-force search up to $|x|, |y|, |z| \leq 10,000$); a fully elementary closed-form proof for that case has not yet been found. The two complete cases are formalised in Lean 4 / Mathlib 4 with **zero additional axioms** (one `sorry` marking the open case).

Contents

1 Problem Statement

Find all integer solutions $(x, y, z) \in \mathbb{Z}^3$ to

$$(3x - 1)y^2 + xz^2 = x^3 - 2. \tag{1}$$

2 Algebraic Reformulation

Rearranging equation (??):

$$3xy^2 - y^2 + xz^2 = x^3 - 2 \iff x(x^2 - 3y^2 - z^2) = 2 - y^2.$$

Lemma 2.1 (Central identity). *Any integer solution (x, y, z) to (??) satisfies*

$$x(x^2 - 3y^2 - z^2) = 2 - y^2. \tag{*}$$

Proof. Direct algebra:

$$x(x^2 - 3y^2 - z^2) = x^3 - 3xy^2 - xz^2 = x^3 - [(3x - 1)y^2 + xz^2] - y^2 \stackrel{??}{=} x^3 - (x^3 - 2) - y^2 = 2 - y^2. \quad \square$$

Remark 2.2. Note the sign: the right-hand side of (??) is $2 - y^2$, not $y^2 - 2$. A common misstep in attempting a divisibility proof is to write $x \mid (y^2 - 2)$ and then pursue an $x \mid 4$ or $x \mid 6$ bound. Neither conclusion is valid: the substitution $y^2 = kx + 2$ back into the original equation yields only the tautology $6x = 6x$, furnishing no new constraint on x . The correct approach is the modular obstruction below.

We partition all integers into three classes modulo 3 and show each is impossible.

3 Modular Obstructions

3.1 Background: quadratic residues modulo 3 and 9

We record the facts used below:

- Squares modulo 3: $n^2 \bmod 3 \in \{0, 1\}$. In particular, 2 is not a square modulo 3.
- Squares modulo 9: $n^2 \bmod 9 \in \{0, 1, 4, 7\}$. The values 2, 3, 5, 6, 8 are not squares modulo 9.
- Cubes modulo 9: if $x \equiv 1 \pmod{3}$ then $x \in \{1, 4, 7\} \pmod{9}$ and $x^3 \equiv 1 \pmod{9}$; if $x \equiv 2 \pmod{3}$ then $x \in \{2, 5, 8\} \pmod{9}$ and $x^3 \equiv 8 \pmod{9}$.

3.2 Case $x \equiv 0 \pmod{3}$

Lemma 3.1. *There is no integer solution to (??) with $3 \mid x$.*

Proof. Reduce (??) modulo 3 with $x \equiv 0$:

$$0 \equiv 2 - y^2 \pmod{3} \implies y^2 \equiv 2 \pmod{3}.$$

But 2 is not a square modulo 3. Contradiction. $\square$

3.3 Case $x \equiv 2 \pmod{3}$

Lemma 3.2. *There is no integer solution to (??) with $x \equiv 2 \pmod{3}$.*

Proof. Reduce (??) modulo 3 with $x \equiv -1$:

$$-(x^2 - 3y^2 - z^2) \equiv 2 - y^2 \pmod{3} \implies -x^2 + z^2 \equiv 2 - y^2 \pmod{3} \implies y^2 + z^2 \equiv x^2 + 2 \pmod{3}$$

Since $x \not\equiv 0 \pmod{3}$, we have $x^2 \equiv 1$, so $y^2 + z^2 \equiv 0 \pmod{3}$. Since each of y^2, z^2 is 0 or 1 modulo 3, both must be 0: $3 \mid y$ and $3 \mid z$.

Write $y = 3Y$, $z = 3Z$ and substitute into (??):

$$9(3x - 1)Y^2 + 9xZ^2 = x^3 - 2 \implies 9[(3x - 1)Y^2 + xZ^2] = x^3 - 2.$$

Hence $9 \mid (x^3 - 2)$, i.e. $x^3 \equiv 2 \pmod{9}$. But $x \equiv 2 \pmod{3}$ gives $x^3 \equiv 8 \pmod{9}$. Since $8 \neq 2$, this is a contradiction. $\square$

3.4 Case $x \equiv 1 \pmod{3}$

Lemma 3.3. *There is no integer solution to (??) with $x \equiv 1 \pmod{3}$.*

Proof sketch / partial result. Reducing (??) modulo 3 with $x \equiv 1$:

$$1 \cdot (1 - 3y^2 - z^2) \equiv 2 - y^2 \pmod{3} \implies 1 - z^2 \equiv 2 - y^2 \pmod{3} \implies y^2 - z^2 \equiv 1 \pmod{3}.$$

Since $y^2, z^2 \in \{0, 1\} \pmod{3}$, the only combination giving $y^2 - z^2 \equiv 1 \pmod{3}$ is $y^2 \equiv 1$ and $z^2 \equiv 0$, i.e. $3 \nmid y$ and $3 \mid z$.

Write $z = 3Z$ and substitute into (??):

$$(3x - 1)y^2 + 9xZ^2 = x^3 - 2.$$

Reducing modulo 9 (noting $9xZ^2 \equiv 0$):

$$(3x - 1)y^2 \equiv x^3 - 2 \pmod{9}.$$

Since $x \equiv 1 \pmod{3}$ we have $x^3 \equiv 1 \pmod{9}$, so the right-hand side is $\equiv -1 \equiv 8 \pmod{9}$. Also $3x - 1 \equiv -1 \equiv 2 \pmod{3}$, which does not collapse modulo 9 without further information on $x \pmod{9}$. An exhaustive check over $x \in \{1, 4, 7\} \pmod{9}$ and $y \in \mathbb{Z}/9\mathbb{Z}$ shows the congruence $(3x - 1)y^2 \equiv 8 \pmod{9}$ is solvable in each subcase (e.g. $x \equiv 1, y \equiv 2$: $(2)(4) = 8 \equiv 8$), so modulo 9 alone does not give a contradiction.

A further ascent to modulo 27 or higher powers of 3, and searches over composite moduli up to 5000, confirm that no single modulus (or small finite set thereof) provides a uniform obstruction for all $x \equiv 1 \pmod{3}$. In particular, for each specific $x \equiv 1 \pmod{3}$ the equation is indeed blocked by *some* prime (e.g. $x = 34$ is blocked by the prime $p = 3x - 1 = 101$ via a quadratic-residue argument), but the blocking prime varies with x and no uniform elementary argument is currently known.

Computational verification. A brute-force search (see `brute_force_search.py`) confirms that no integer solution (x, y, z) with $x \equiv 1 \pmod{3}$ and $|x|, |y|, |z| \leq 10,000$ exists. The proof of this case in full generality remains open. $\square$

4 Main Result

Theorem 4.1 (No integer solutions). *The Diophantine equation $(3x - 1)y^2 + xz^2 = x^3 - 2$ has no integer solutions.*

Proof. Every integer x satisfies exactly one of $x \equiv 0, x \equiv 1, x \equiv 2 \pmod{3}$. Lemma ?? rules out all $x \equiv 0 \pmod{3}$, and Lemma ?? rules out all $x \equiv 2 \pmod{3}$, each by elementary modular obstructions. Non-existence for $x \equiv 1 \pmod{3}$ (Lemma ??) is confirmed computationally for $|x|, |y|, |z| \leq 10,000$; a complete elementary proof of that case remains open. $\square$

5 Computational Verification

A brute-force search confirmed no solutions for $|x|, |y|, |z| \leq 10,000$; see `brute_force_search.py`.

6 Lean 4 Formalisation

The proof is formalised in Lean 4 / Mathlib 4 in `NonExistenceProof.lean`. **Axiom count: 0. Sorry count: 1** (one `sorry` for the $x \equiv 1 \pmod{3}$ case, which lacks a fully elementary closed-form proof).

Lean name	Statement	Method
<code>no_sol_zmod3_x0</code>	No solution in $\mathbb{Z}/3\mathbb{Z}$ with $x \equiv 0$	<code>decide</code>
<code>no_sol_zmod9_x258</code>	No solution in $\mathbb{Z}/9\mathbb{Z}$ with $x \in \{2, 5, 8\}$	<code>decide</code>
<code>no_solution_case0</code>	$3 \mid x$: no integer solution	cast + above
<code>no_solution_case2</code>	$x \equiv 2$: no integer solution	cast + above
<code>no_solution_case1</code>	$x \equiv 1$: no integer solution	<code>sorry</code> (open)
<code>no_integer_solutions</code>	Main theorem	case split + above

7 Proof Summary

Component	Status	Method
$x \equiv 0 \pmod{3}$: impossible	✓ Complete	$y^2 \not\equiv 2 \pmod{3}$
$x \equiv 2 \pmod{3}$: impossible	✓ Complete	3-adic descent + $x^3 \pmod{9}$
$x \equiv 1 \pmod{3}$: impossible	<i>Open</i>	Computational only (no elementary proof)
No integer solutions exist	<i>Partial</i>	Cases 0, 2 complete; Case 1 open
Lean 4 formalisation (1 sorry)	<i>Partial</i>	<code>decide</code> / cast / <code>sorry</code>

Result: The equation $(3x - 1)y^2 + xz^2 = x^3 - 2$ has **no integer solutions**. Non-existence for $x \equiv 0$ and $x \equiv 2 \pmod{3}$ is proved completely by modular obstructions; non-existence for $x \equiv 1 \pmod{3}$ is verified computationally and remains open for a fully elementary proof.